

Equality of Permutations

Let f and g be two permutations on a set X .

Then $f = g$ if and only if $f(x) = g(x)$ for all x in X .

Ex

Let f and g be given by

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad g = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix}$$

Evidently

$$f(1) = 2 = g(1)$$

$$f(2) = 3 = g(2)$$

$$f(3) = 4 = g(3)$$

$$f(4) = 1 = g(4)$$

Then

$$f(x) = g(x) \text{ for all } x \in \{1, 2, 3, 4\}$$

which implies $f = g$

Identity Permutation

If each element of a permutation is replaced by itself. Then it is called the identity permutation and is denoted by the symbol I . For, on -

$$I = \begin{pmatrix} a & b & c \\ a & b & c \end{pmatrix} \text{ is an identity permutation,}$$

Product of Permutation / Composition

$$f = \begin{pmatrix} a_1 a_2 a_3 \dots a_n \\ b_1 b_2 b_3 \dots b_n \end{pmatrix}$$

$$g = \begin{pmatrix} b_1 b_2 b_3 \dots b_n \\ c_1 c_2 c_3 \dots c_n \end{pmatrix}$$

$$f \circ g = \begin{pmatrix} a_1 a_2 a_3 \dots a_n \\ c_1 c_2 c_3 \dots c_n \end{pmatrix}$$

Find the product of two permutations and

Show that it is not commutative.

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix}$$

$$g = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 4 \end{pmatrix}$$

$$fg = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 4 \end{pmatrix}$$

(3)

$$= \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \begin{pmatrix} 2 & 1 & 4 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$$

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 4 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 4 \end{pmatrix} \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix}$$

observe that $fg \neq gf$.

This shows that the product of two permutations is not commutative.

Normal Subgroup

(4)

Let G be a group and H is a subgroup under the same binary operation $*$, then for any $a \in G$,

$$aH = \{a * h \mid h \in H\}$$

$$Ha = \{h * a \mid h \in H\}$$

$H \trianglelefteq G$ {Representation}

then,

$$aH = Ha \quad \{\text{Left Coset} = \text{Right Coset}\}$$

that subgroup H is called as Normal subgroup of G .

For abelian group - $aH = Ha$ $H = \text{Normal subgroup}$
 $\forall a \in G$

Improper/Trivial Normal subgroup - $\begin{cases} \{e\} & ea = ae \\ & \forall a \in G \\ \{G\} & Ga = aG \end{cases}$
 $Ga = aG \quad \forall a \in G$

Each group has two normal subgroups - $\{e\}$ and $\{G\}$

Other than $\{e\}$ and $\{G\}$ normal subgroups, are called proper normal subgroups.

Q Let H be a subgroup of the group G . then H is a normal subgroup iff $aha^{-1} \in H \quad \forall a \in G, h \in H$

Let H is a normal subgroup i.e. $aH = Ha$

we have to prove that $aha^{-1} \in H$

let $x \in aH$, $x \in Ha$

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$$x = a * h - (1)$$

$$x = h_1 * a - (2)$$

from (1) and (2) $a * h = h_1 * a$

Multiply a^{-1} on L.H.S and R.H.S

$$\begin{aligned} (a * h) * a^{-1} &= (h_1 * a) * a^{-1} \\ &= h_1 * (a * a^{-1}) \\ &= h_1 * e \end{aligned}$$

$$(a * h) * a^{-1} = h_1 \quad \& \quad h_1 \in H$$

$$\therefore (a * h) * a^{-1} \in H$$

$$\text{or } aha^{-1} \in H \quad \forall h \in H, a \in G$$

let $aha^{-1} \in H$ and we have to prove that H is normal subgroup i.e. $aH = Ha$.

$$\text{let } x \in aH - (1)$$

$$x = a * h = ah$$

Multiply a^{-1} on both sides

$$xa^{-1} = aha^{-1}, \text{ here } aha^{-1} \in H \text{ (acc. to assumption)}$$

So,

$$xa^{-1} \in H$$

Multiply a on both sides

$$(xa^{-1})a \in Ha$$

$$x(a^{-1}a) \in Ha$$

$$xe \in Ha$$

$$x \in Ha - (2)$$

from (1) and (2) $aH \subseteq Ha$

Similarly we can prove that $Ha \subseteq aH$
Hence,

$$aH = Ha$$

(I) & (II)

So H is a normal subgroup iff
 $aH a^{-1} \in H \quad \forall \quad a \in G, h \in H.$

FACTOR / QUOTIENT GROUP

A group whose elements are the cosets of a normal group of a given group is called a quotient or factor group.

If G is a group and N is a Normal subgroup of G , then the set of all cosets of N in G forms a group. This group is called as quotient group, or factor group of G relative to N .

It is denoted by G/N .

$$G/N = \{ Na \mid a \in G \} \text{ or}$$

$$\{ aN \mid a \in G \}$$

$$\text{as } aN = Na$$

Theorem :- Self